Greenness and PM2.5 Across Shenzhen Districts

Research question

This report asks a simple question: do districts with more greenery also tend to have lower PM2.5? I compare all ten Shenzhen districts in 2025. This is a comparison of patterns, not a test of cause and effect.

Data and measures

The PM2.5 data are daily district averages from the Shenzhen Open Data Platform (dataset 29200_01003608). I compare the 2025 yearly averages. District boundaries come from the course files. For greenery, I use 2025 Sentinel-2 satellite images in Google Earth Engine (COPERNICUS/S2_SR_HARMONIZED).

I use NDVI, a satellite score that shows how green a surface is. Higher NDVI usually means more vegetation. I remove clouds, shadows, water, and other unusable pixels. I also use a second vegetation score based on an NDVI cutoff of 0.30. The score is summarized for each district as a percentage. These are estimates, not official green-cover percentages. They include all types of vegetation, not only parks, and do not show how much green space residents can access.

What the data show

The vegetation score ranged from 49.5% in Longhua to 87.2% in Dapeng. NDVI was also lowest in Longhua (0.379) and highest in Dapeng (0.711). Mean PM2.5 was highest in Longhua $18.82 \mu g/m^3$ and lowest in Dapeng $14.00 \mu g/m^3$.

The rank-based correlation was -0.879 $p = 0.002$, a strong negative pattern: districts with higher vegetation scores tended to have lower PM2.5. Because this is a small, observational comparison, it cannot show that vegetation caused lower PM2.5.

District results

The table gives the values for each district. “Days” shows how many daily PM2.5 records were available out of 365.

District	Mean NDVI	Vegetation indicator (%)	Mean PM2.5 $\mu g/m^3$	Days
Dapeng	0.711	87.2	14.00	365
Luohu	0.550	70.7	14.96	365
Yantian	0.569	70.7	14.94	365
Pingshan	0.532	69.5	15.46	365
Nanshan	0.463	62.4	17.93	365
Guangming	0.439	61.1	17.53	365
Longgang	0.433	57.5	17.89	365
Futian	0.410	55.9	16.54	365
Baoan	0.384	50.2	18.18	364
Longhua	0.379	49.5	18.82	365

Table 1. Greenness and PM2.5 by Shenzhen district, 2025.

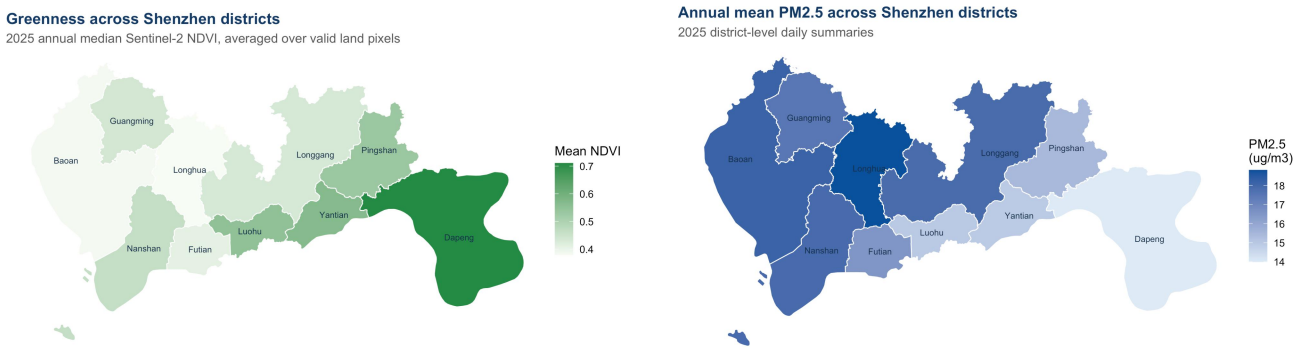


Figure 1. Maps of average NDVI (left) and average PM2.5 (right) in 2025.

Greenness and PM2.5

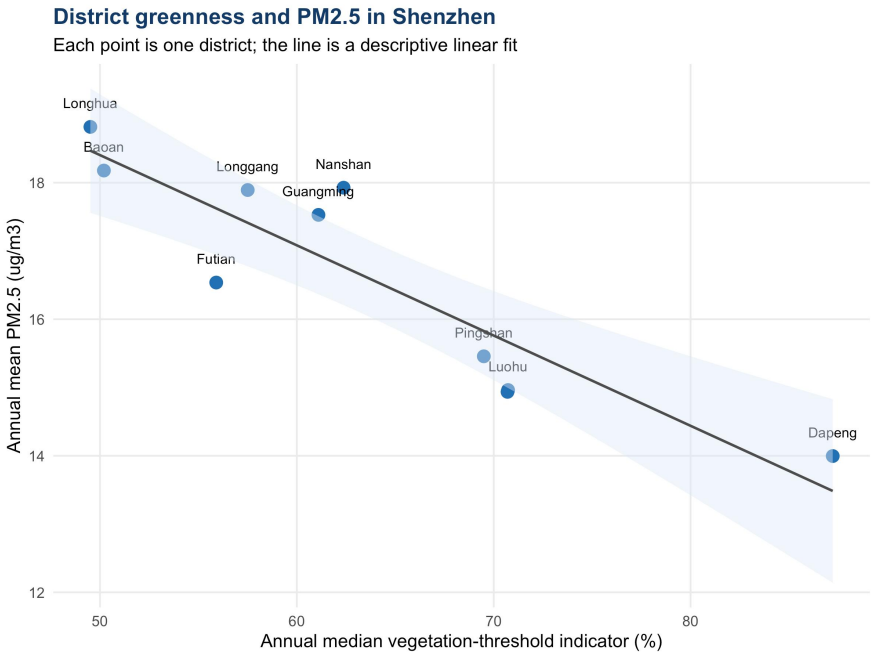


Figure 2. Each dot is one district. The downward line shows the overall pattern, but does not prove cause and effect.

What to keep in mind

This analysis has only ten districts, so I treat the result as an early pattern. Dapeng is both the greenest district by our vegetation score and the district with the lowest PM2.5. Without Dapeng, the relationship remained negative (correlation = -0.833, p = 0.0083). This suggests Dapeng alone does not explain the pattern, but the sample is still small.

NDVI includes forests and hills, not only public parks. PM2.5 can also be affected by traffic, industry, weather, and monitor locations. We have district summaries, not readings from every monitor. Baoan has one missing day (99.7% coverage), and I did not fill it in. Because this is a small comparison, I cannot conclude that greenery reduces PM2.5.